Thales CTE Agent for Linux — Upgrade

Standard Operating Procedure (SOP) — Thales CipherTrust Transparent Encryption (CTE) Agent for Linux

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Engineer Session Cheat Sheet (Linux Commands)

Quick copy/paste reference for active engineering work sessions.

Use this as a convenience block to copy commands into working notes or terminal workflow.

Important: Engineers should still read the full procedure document end-to-end to understand the purpose, prerequisites, timing, risks, validation steps, and when each command should or should not be used.

```
# --- Pre-Requisites & Environment Setup ---
vmsec version
mount /tjx/SW_depot
cd /tjx/SW_depot/CipherTrust

# --- Application & Session Quiescence ---
lsof +D /guardpoint_directory
fuser -cv /guardpoint_directory

# --- Stop and Uninstall Legacy Agent ---
/etc/vormetric/secfs stop
/opt/vormetric/DataSecurityExpert/agent/vmd/bin/uninstall

# >>> REBOOT THE SERVER IF "device is busy" ERRORS PERSIST <<<

# --- Prepare New Binary ---
sudo dnf install -y libns12
chown root:root vee-fs-<version>.bin
chmod 755 vee-fs-<version>.bin
mount -o remount,rw,exec /tmp

# --- Install New Agent ---
./vee-fs-<new-version>.bin

# >>> REBOOT THE SERVER AFTER INSTALL <<<

# --- Post-Reboot Verification ---
/etc/vormetric/secfs status
vmsec version
secfsd -status guard
```

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Reference	Standard Value / Location
Binary Source Repository	`/tjx/SW_depot/CipherTrust`
Primary Management Server (CTM)	`zctm001p.corp.tjxcorp.net`
Technical Support	EDS Data Security Team (mlcyber_data_security_engineering@tjx.com)

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1. Executive Summary and Scope

This document serves as the authoritative Standard Operating Procedure (SOP) for the AHS-Unix-L2 Engineering and Patching teams. It details the precise workflow required to **upgrade and verify** the Thales CTE Agent on Linux Server infrastructure (e.g., RHEL 7/8/9).

Operational Objective: Ensure the secure upgrade of the CTE Agent to facilitate "Encryption as a Service" without disrupting business-critical applications.

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2. Deployment Pre-requisites and Environment

> **Critical Warning:** Failure to validate these pre-requisites prior to execution will result in installation failure or server instability.

2.1 Change Management Validation

Before initiating work, verify the ServiceNow Change Request (CTASK):

- **Version Control:** Confirm the exact version string required (e.g., `vee-fs-7.4.0-154-rh7-x86_64.bin`).

2.2 System State and Kernel Validation

Linux CTE Agents require strict kernel compatibility.

1. **Check Current Version:** Run `vmsec version` to capture the current state before upgrading.
2. **Kernel Check:** Ensure the OS kernel matches the supported versions for the CTE binary being installed.
3. **Backup Verification:** Verify that the protected data has a **FULL Last Known Good** backup prior to upgrading the agent software.

2.3 Binary Source Retrieval

1. **Mount Depot:** Ensure the software depot is mounted to the target server:

```
mount /tjx/SW_depot
```

2. **Navigate:**

```
cd /tjx/SW_depot/CipherTrust
```

3. **Copy Binary:** Copy the specific `.bin` file to the local `/tmp` directory.

> **Constraint:** Never run the installer directly from the network mount to prevent latency-induced corruption.

2.4 OS Dependency Validation

The CTE Agent requires specific 64-bit libraries to function correctly. The AHS-Unix-L2 Engineer may encounter a dependency error during installation if the `libns12` library is missing.

Action: Ensure the **64-bit** version of `libns12` is installed prior to execution. Do **not** install the 32-bit or i686 version.

Command:

```
sudo dnf install -y libns12
```

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3. Upgrade Lifecycle Management

Use this procedure for Linux servers with an **existing** agent (e.g., patching from v7.4.x to v7.5.x).

For Linux, the standard upgrade protocol involves **cleanly uninstalling the old agent and installing the new one**. This is not an in-place upgrade — it is a full remove-and-reinstall cycle.

3.1 Upgrade Execution Protocol

> **Critical Step: Handle & Session Release**
>
> Upgrading while files are open or in-use by users/applications will cause the uninstaller to fail or corrupt the kernel driver state. All file handles to protected GuardPoints must be closed.

Step 1: Quiesce Applications (Stop Services)

Identify the primary workload (e.g., Oracle, DB2, Talend).

Action: Stop these application streams gracefully.

Step 2: Verify and Close Open File Handles

> **Warning:** Commands like `fuser` and `lsof` might not always reveal an active GuardPoint because they detect active usage, not locked states.

```
lsof +D /guardpoint_directory
fuser -cv /guardpoint_directory
```

If you suspect files are still locked, or if an uninstall attempt returns a `/path: device is busy` error, the safest operational step is to **reboot the server from the VM console** to clear any hung processes before proceeding with the upgrade.

Step 3: Stop and Uninstall Legacy Agent

1. **Stop** the active CTE services:

```
/etc/vormetric/secfs stop
```

2. **Execute** the uninstall script:

```
/opt/vormetric/DataSecurityExpert/agent/vmd/bin/uninstall
```

3. Wait for the **"Success!"** message indicating the agent was cleanly removed.

Step 4: Execute New Installation

1. **Prepare** the new binary in `/tmp` (ensure `chmod 755` and `remount,exec` as detailed in Section 2.3):

```
chown root:root vee-fs-<version>.bin
chmod 755 vee-fs-<version>.bin
mount -o remount,rw,exec /tmp
```

2. **Verify** the `libns12` dependency is met as detailed in Section 2.4.
3. **Launch** the new binary:

```
./vee-fs-<new-version>.bin
```

4. Follow the **Registration Configuration** (Section 3.2) to re-register the agent using the provided Token and answering the wizard prompts exactly as listed.

3.2 Registration Configuration (Re-Registration)

Because the upgrade process is a full uninstall and reinstall, the agent must be re-registered with the CTM. Use the appropriate **Registration Token** below to authenticate the agent for its specific environment.

Registration Tokens

Copy the appropriate string from the boxes below and paste it when prompted for the **"Registration Token"** in the wizard:

Production Token:

```
bAlAUI2y6KEh1OMJNbuDEbH59yklzWkI53L8kv3Tnzjq2GZOzXWojCGKxzSlzjvq
```

Non-Production Token:

```
bbaoayRBfZtwrozobysX8IxNbH7VaPZjgRGT0OzrqkLjoTfDaVh0NMlu7niDu0mZ
```

Wizard Logic & Parameters Table

Navigate the interactive prompts using the exact logic below:

Prompt / Screen	Required Action / Configuration Value
License Agreement	Press **Enter** to scroll (be careful near 99%). Type **Y** to accept.
Continue with agent registration?	Type **Y**
Primary key manager host name	`zctm001p.corp.tjxcorp.net`
Is this host name correct?	Type **Y**
What is the name of this machine?	Select the number corresponding to the **Full FQDN** (e.g., `[1] myserver.tjx.com`).

Registration Token	Paste the specific **Production** or **Non-Production** token from the boxes above.
Profile Name	**Non-Production:** Enter `DefaultProfile`
	Production: Enter `DefaultClientProfile`
Host Group Name	*(Leave blank / Press Enter)*
Host description	*(Leave blank / Press Enter)*
Are the above values correct?	Type **Y**
Do you want to enable this functionality? *(Hardware association prompt)*	Type **N** *(Required to support VM mobility/restores)*
Do you want this host to have Filesystem encryption support enabled on the server?	Type **Y**
Do you want this host to have File Activity Monitoring support enabled?	Type **N**
Do you want this host to have LDT support enabled on the server?	Type **N**
Do you want to configure this host for Cloud Object Storage?	Type **N**
Do you want this host to have Ransomware protection support enabled on the server?	Type **N**

3.3 Post-Upgrade Sequence

1. **Mandatory Action:** Reboot the server. This ensures all hung handles are cleared and the new driver attaches to the data mounts cleanly.

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4. Operational Verification and Auditing

Upon reboot, the AHS-Unix-L2 engineer must validate the agent is active and healthy using the following tiered approach.

4.1 Tier 1: Service Layer Verification

Open a root terminal and check the daemon status:

```
/etc/vormetric/secfs status
```

Expected Output: Services should show as **active (running)**.

4.2 Tier 2: Version and Connectivity Verification

Execute the standard version check to confirm the new binary is active and communicating with the CTM:

```
vmsec version
```

Expected Output: Displays the new CTE version number and confirms connection.

4.3 Tier 3: Deep Technical Verification (Guard Points)

Check the physical enforcement of the encryption policies:

```
secfsd -status guard
```

Expected Output: The output should list all protected directories/devices with a status indicating they are **active** and **guarded**.

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5. Troubleshooting and Error Handling

Error / Symptom	Root Cause	Resolution
"Permission denied" when running `.bin`	The `/tmp` directory is mounted with the `noexec` flag.	Run `mount -o remount,rw,exec /tmp` before executing the binary.
Missing `libns12` dependency error	The 64-bit library required by CTE is not present.	Run `sudo dnf install -y libns12`. Do **not** install the 32-bit version.
"/path: device is busy" during Uninstall	A database or application is maintaining a locked state on the GuardPoint.	1. Use `lsof` to find active PIDs. 2. If it persists, reboot the server to clear hung file handles and retry the uninstall.
Registration Timeout / Cannot reach CTM	Network firewall blocking Port 443 outbound to the CTM.	Run `curl -v <https://zctm001p.corp.tjxcorp.net:443`.> If it

		fails, contact the EDS Data Security team.
"Unauthorized" / "Failed to Register"	Invalid Token entered during the wizard prompt.	Verify the proper Prod/Non-Prod Token from Section 3.2 was copied exactly, with no leading or trailing spaces.

Important:

- If a backout plan or rollback occurs, please open a ServiceNow incident, assign it to SEC-CYBER DATA SECURITY ENGINEERING-L4, and route it through the MIM process to ensure immediate engagement of the on-call Data Security Engineer